PERSISTENT COEFFICIENT VANISHING IN QUINTUPLE PRODUCTS: ADMISSIBLE TRIPLES AND A PRIME FOUR-FACTOR FAMILY

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Abstract. For a prime modulus and three distinct canonical indices whose squared sum is divisible by that modulus, we give a necessary and sufficient arithmetic criterion for an entire coefficient progression of a product of three quintuple products to vanish. The criterion is a short vector in a projective dual lattice, together with an explicit residue equation. The equivalence is proved in Lean. We explain the analytic and lattice steps, the finite certificates below the geometric threshold, and a linear scan of projective scalars that implements the criterion. We also prove a necessary and sufficient criterion for the four-factor reflection family at prime moduli that are sums of two squares, and give the optimal nonvanishing cutoff for its modulus-97 example. This is a manuscript for independent mathematical and priority review. Arbitrary four-factor tuples and composite-modulus necessity remain outside its scope.

1. Statement and scope

Write $(a;q)_\infty = \prod_{n\geq 0}(1-aq^n)$ and fix the convention

$$Q(z,q) = (z, q/z, q; q)_\infty (qz^2, q/z^2; q^2)_\infty.$$

Thus $Q(q^i, q^p)$ is an integral formal power series when $0 < 2i < p$. Put

$$F_{p,v}(q) = Q(q^i, q^p)Q(q^j, q^p)Q(q^k, q^p) = \sum_{n\geq 0} a_{p,v}(n)q^n, \qquad v = (i,j,k).$$

Definition 1. The triple $(p; i, j, k)$ is *admissible* if p is prime, $p \neq 3$, $0 < i < j < k$, $2k < p$, and $p \mid i^2 + j^2 + k^2$.

Theorem 2 (Admissible triple classification)**.** *Suppose $(p; i, j, k)$ is admissible and $0 \leq R < p$. Then $a_{p,v}(pN + R) = 0$ for every $N \geq 0$ if and only if there are $w \in \mathbb{Z}^3$, $\lambda \in \mathbb{Z}$ and $e \in \{1, 2\}$ such that*

$$w \equiv \lambda v \pmod p, \qquad \|w\|^2 = ep,$$

and, on setting $u = v \cdot w/p$ and $c = 2/e$, one has

$$2\lambda R \equiv w_1 + w_2 + w_3 - 6u \pmod p, \qquad ep \equiv 1 \pmod 3, \tag{1}$$

$$c\lambda(2\lambda R - w_1 - w_2 - w_3 + 6u) \equiv 12 \pmod p, \qquad ep \equiv 2 \pmod 3. \tag{2}$$

Here u is an integer by isotropy, and λ is automatically nonzero modulo p. Indeed, $w \in p\mathbb{Z}^3$ and $\|w\|^2 = ep$ would imply $p \mid e$, which is impossible. The residue equation is part of the criterion: existence of a short root alone does not determine R.

The exact formal endpoint is `admissible_root_vanishing_equivalence` in `RamanujanTau/MultiQuintupleFullConverse.lean`. Its hypotheses are precisely admissibility and $R < p$. Its vanishing predicate refers to the coefficients of the actual infinite products, rather than a truncated series or a hypothesized analytic identity.

The prime and distinct-index assumptions are material. For example, common scaling at composite modulus can produce automatic support vanishings with no root of the stated kind. The repository proves such an obstruction at modulus 9 with three indices equal to 3. No claim about all triples, composite moduli, or repeated indices is made here.

Date: Review draft, November 12, 2026.

2. Relation to the literature

Daniels, Huber, McLaughlin and Ye study two-factor products by dissection and lattice involutions [1, 2]. Their concluding sections discuss higher-factor examples; in particular, the second paper lists the triple $(31; 1, 5, 6)$ and the quadruple $(41; 1, 4, 5, 9)$ as experimental examples, and announces further work. The theorem above addresses an entire admissible class and includes necessity. Their earlier discussion also records an already proved triple example at modulus 13; individual triple identities are therefore not claimed as a new phenomenon.

Chern and Tang give general coefficient-vanishing methods using theta series and huffing operators [3]. The distinction requiring specialist review is whether the exact necessary-and-sufficient criterion of Theorem 2, under its precise admissibility conditions, already follows from that work or subsequent literature. Searches and comparisons recorded on November 11, 2026 did not identify that exact classification. This is evidence for further review, not a priority certification. The present draft makes no "first proof" claim.

3. Proof of the classification

We give the argument with the intermediate arithmetic exposed. The linked formal modules provide the complete details of convergence, summation, lattice generation and the exceptional finite range.

3.1. **From coefficients to dual shells.** Watson's identity writes one factor as

$$Q(q^i, q^p) = \sum_{n\in\mathbb{Z}} (q^{A_i(n)} - q^{B_i(n)}), \quad A_i(n) = \frac{pn(3n-1)}{2} + 3in, \quad B_i(n) = A_i(n) + i + pn.$$

All these exponents are nonnegative for canonical indices. For a branch $b \in \{0,1\}^3$, the progression condition becomes $3v \cdot n + v \cdot b \equiv R \pmod p$. Define

$$L = \{x \in \mathbb{Z}^3 : v \cdot x \equiv 0 \pmod p\}, \quad \mathcal{N} = pL^* = \{w \in \mathbb{Z}^3 : w \equiv \lambda v \pmod p\}.$$

Their covolumes are p and p^2. Choose q_0 with $3kq_0 \equiv 1 \pmod p$, and set

$$c_b \equiv q_0(R - v \cdot b) \pmod p, \qquad x_b = \frac{v}{p} + \frac{2b - \mathbf{1}}{6} + c_b(0, 0, 1).$$

Completing squares and applying Poisson summation to the eight Gaussian coset sums shows that persistent vanishing forces the amplitude sum on each norm shell of $\mathcal{N}$ to be zero. All sums involved are absolutely convergent at positive Gaussian parameter.

For completeness, let $\zeta = e^{2\pi i/(6p)}$ and $u = v \cdot w/p$. The dual amplitude is

$$A_R(w) = \zeta^{C(w)} \prod_{s=1}^{3} (1 - \zeta^{\delta_s(w)}), \quad C(w) = 6w_3q_0R + 6u - \sum_s w_s, \quad \delta_s(w) = 2w_s - 6w_3q_0v_s.$$

It is nonzero exactly when every coordinate of w is nonzero modulo 3; call such a vector supported. Write $t_s = \delta_s(w)/(2p) \pmod 3$ and let $n(w)$ be the number of t_s equal to 2. For supported w,

$$A_R(w) = (1 - \zeta^{2p})^3 \rho^{a(w)}, \qquad \rho = \zeta^3, \qquad a(w) = \frac{C(w) + pn(w)}{3} \pmod{2p}.$$

The numerator is divisible by 3, and $a(-w) = p - a(w)$. The cyclotomic relation for a primitive $2p$th root, combined with this antipodal identity, forces equal multiplicities at phases a and $a + p$. Consequently every supported vector has an equal-norm opposite-phase partner. This argument does not impose a bound on the size of a shell.

3.2. **A coherent pairing below the short-vector threshold.** Work modulo p, and define

$$h = \frac{v \cdot v}{p}, \quad c_0 = \frac{2R - \sum v_s}{3}, \quad D = h + c_0, \quad \ell(w) = 2\frac{v \cdot w}{p} + c_0\lambda(w).$$

The scalar $\lambda(w)$ is uniquely determined modulo p. Expanding $w = \lambda v + pz$ gives

$$\|w\|^2/p = \lambda(w)\ell(w) - D\lambda(w)^2, \tag{3}$$

$$2w \cdot z/p = \lambda(w)\ell(z) + \lambda(z)\ell(w) - 2D\lambda(w)\lambda(z). \tag{4}$$

All pairings in $\mathcal{N}$ are divisible by p.

Lemma 3 (Short-vector uniqueness)**.** *If distinct integral vectors w, w' have the same norm and are congruent modulo an odd integer p, then $\|w\|^2 \geq p^2/2$.*

Proof. Write $w' - w = pz$. Equality of norms gives $2w \cdot z + p\|z\|^2 = 0$, so $\|z\|^2$ is a positive even integer. The parallelogram identity gives $p^2\|z\|^2 \leq 4\|w\|^2$. □

Let S consist of supported vectors of norm squared less than $p^2/2$. For opposite partners $w, w' \in S$, the phase condition preserves ℓ modulo p. Their scalars are different by the lemma. Subtracting (3) gives

$$D(\lambda(w) + \lambda(w')) = \ell(w).$$

If S is nonempty then $D \neq 0$: otherwise $\ell(w) = 0$ and $\|w\|^2/p \equiv 0 \pmod p$, contradicting $0 < \|w\|^2 < p^2/2$. Thus partners are unique. Equation (4) shows that the partner permutation preserves dot products modulo p^2. Cauchy–Schwarz bounds the absolute difference of the two dot products strictly below p^2, so it preserves them as integers.

3.3. **Integral generation, uniformly in the prime.** For $p \neq 13$, S generates $\mathcal{N}$ over $\mathbb{Z}$. Here is the quantitative part of that assertion. The integral lattice with underlying module $\mathcal{N}$ and inner product $(w \cdot z)/p$ has Gram determinant p. Its adjugate lattice is identified with L, whose minimum squared norm is at least 3: norm 1 would force a zero index modulo p, and norm 2 would force two indices equal up to sign.

A reduced basis has Gram–Schmidt squared lengths A, B, C and coefficients μ, ν, ξ satisfying

$$|\mu|, |\nu|, |\xi| \leq \tfrac{1}{2}, \quad ABC = p, \quad B \geq \tfrac{3}{4}A, \quad C \geq \tfrac{3}{4}B, \quad A \in \mathbb{Z}_{>0}, \quad M = AB \in \mathbb{Z}_{\geq 3}.$$

Take coefficient vectors (x, y, z) with $x, y \in \{-2, -1, 0, 1, 2\}$, $z \in \{0, 1\}$, which are nonzero isotropic vectors for the Gram form modulo 3. Their images are supported and generate $\mathcal{N}$. Indeed, reduction to $\mathbb{F}_3^3$ is an isomorphism, a nonzero isotropic vector for a sum of three squares has all coordinates nonzero, and the indicated box generates integrally: its reductions span, and differences of representatives produce $3\mathbb{Z}^3$. Their scaled squared norms are bounded by

$$C + \frac{49A + 25B}{4} = \frac{p}{M} + \frac{49A + 25B}{4}.$$

If $M < 12$, then $A \leq 3$, $B \leq 11$, giving $p/3 + 211/2 < p/2$ for $p \geq 1250$. If $M \geq 12$, the reduction inequalities give $A^3 \leq 64p/27$ and $B^2 \leq 4p/3$; hence $A \leq p/80$ and $B \leq p/30$ for $p \geq 1250$. The bound is then $427p/960 < p/2$. Returning to the unscaled norm gives the required strict bound $p^2/2$.

Below 1250 there are 14,370 normalized canonical triples at 202 primes. For 14,369 of them the supplied certificate consists of at most five supported short vectors whose 3×3 minors have gcd p^2. That gcd proves integral generation, rather than rational spanning. Normalization, signed coordinate permutations and projective rescaling cover every admissible triple in this range. The exception is modulus 13. There the two canonical triples are $(1, 3, 4)$ and $(2, 5, 6)$. Their zero residues are respectively $\{2, 4, 10\}$ and $\{3, 9, 11\}$; six root certificates and twenty explicit nonzero coefficients settle both directions. All finite certificates and the coverage argument are checked in Lean. An independent integer-only checker is also supplied.

3.4. **Extracting the root and its residue.** For $p \neq 13$, the dot-product-preserving permutation of the integral generating set S extends uniquely to a rational orthogonal involution T, with $T(\mathcal{N}) = \mathcal{N}$ and hence $T(L) = L$. It is neither I nor $-I$. The first is excluded by the opposite-phase condition. The second would force ℓ to vanish on $\mathcal{N}$, whereas $\ell(pe_s) = 2v_s \neq 0 \pmod p$. In dimension three, T is a reflection H_w or its negative, for a primitive integral normal w.

Set $n = w \cdot w$. Integrality on $pe_s \in L$ gives $n \mid 2pw_sw_t$ for every s, t, and primitivity implies $n \mid 2p$. The possibilities $n = 1, 2$ contradict admissibility. Thus $n = ep$, $e \in \{1, 2\}$; integrality on L then gives $w \equiv \lambda v \pmod p$.

The phase parity on a supported vector x is

$$(-1)^{a(x)} = (-1)^{\sum x_s} \chi_3(p)\chi_3(x_1x_2x_3).$$

Every orthogonal matrix for a sum of three squares over $\mathbb{F}_3$ is a signed permutation. Applying this to T selects $T = H_w$ when $ep \equiv 1 \pmod 3$ and $T = -H_w$ when $ep \equiv 2 \pmod 3$. With $c = 2/e$ and $t_x = x \cdot w/p$, the reflection is $H_w(x) = x - ct_xw$. Preservation of ℓ gives $\ell(w) = 0$ in the first case and $c\lambda\ell(w) = 4$ in the second. Since $3\ell(w) = 6u + \lambda(2R - \sum v_s)$, these are exactly (1) and (2).

Conversely, a root with the indicated residue equation induces the sign-reversing involution on the Watson branches, preserving the exponent in every coefficient sum. Pairwise cancellation proves persistent vanishing. This direction is the formal theorem `hasProjectiveRootTarget_implies_persistentTripleVanishing`. Together with the exceptional computation at 13, this proves Theorem 2.

4. An exact finite decision procedure

Proposition 4. *For an admissible triple, all vanishing residues can be found by scanning the $p-1$ nonzero scalars modulo p. For each scalar λ, take the centered representative $w_s \in [-(p-1)/2, (p-1)/2]$ of λv_s. Retain w if its squared norm is p or $2p$, and solve the applicable residue equation for R. The resulting set is exactly the set of persistent zero progressions.*

Proof. Admissibility implies $p \geq 7$. If $\|w\|^2 \leq 2p$, every integral coordinate satisfies $|w_s| \leq (p-1)/2$, since $((p+1)/2)^2 > 2p$ for $p \geq 7$. Thus every root in Theorem 2 is the centered representative for its scalar. Conversely, every retained vector satisfies the root conditions. The coefficients 2λ and $2c\lambda^2$ in the two equations are units modulo p, so each root gives one residue. Apply the theorem. □

This takes $O(p)$ arithmetic operations when inverses of all nonzero residues are precomputed, and $O(p \log p)$ using a separate extended Euclidean inversion for each retained root. This counts arithmetic operations, not bit operations. No coefficient truncation is needed to decide persistent vanishing. The supplied Python implementation uses exact integers and modular inverses; it is a reference implementation, not extracted Lean code. The completeness ingredient that every certified root is the centered representative for its scalar is also checked in Lean as `projectiveRootTarget_centered`. The executable's full input/output correspondence is the elementary argument above, rather than a code-extraction or refinement theorem.

5. The prime four-factor reflection family

For natural numbers m, n, put $p = m^2 + n^2$ and

$$G_{m,n}(q) = Q(q^{m-n}, q^p)Q(q^n, q^p)Q(q^m, q^p)Q(q^{m+n}, q^p) = \sum_{r \geq 0} g_{m,n}(r)q^r.$$

Theorem 5 (Prime reflection-family classification)**.** *Suppose p is prime, $0 < n < m$, $2(m+n) < p$, and $0 \le R < p$. Then $g_{m,n}(pN + R) = 0$ for every $N \ge 0$ if and only if*

$$9 \mid m \quad or \quad 9 \mid n \quad or \quad 9 \mid (m-n) \quad or \quad 9 \mid (m+n),$$

and at least one of the following congruences holds modulo p:

$$2R \equiv 3m + n - 18, \qquad 2R \equiv 3m + n,$$
$$(n-m)R \equiv n(m+2n-9), \qquad (m+n)R \equiv n(2m-n-9).$$

The formal endpoint is `coeff_fourReflection_prime_iff` in `FourPrimeConverse.lean`. There is no additional spanning assumption and no excluded small prime. The divisibility condition and the residue condition are both necessary.

5.1. **Proof architecture.** Sufficiency follows from explicit sign-reversing affine involutions on the sixteen Watson branches. For necessity, the four-dimensional Poisson bridge turns persistent coefficient vanishing into cancellation on shells of the projective dual lattice. At norm below $p^2/2$, the $p-1$ nonzero projective scalars leave a Fourier phase unrepresented. The missing phase forces cancellation to pair each supported vector with an equal-norm opposite-phase vector, reversing its modulo-three quartic character. This replaces the odd-dimensional antipodal argument used for triples.

Integer generator certificates, checked in Lean, promote those partners to an orthogonal involution preserving both relevant lattices. Preservation of a distinguished short plane and its orthogonal long plane reduces the map to two signed-permutation blocks. An exhaustive modulo-nine calculation excludes the incompatible blocks and leaves precisely the divisibility condition above. Phase preservation then yields the four residue equations. The finite range is completed by actual-product coefficient certificates at $13, 17, 29, 37, 61, 73, 89$ and the existing classifications at $41, 53, 97$. The source map below separates these analytic, integral and finite steps.

5.2. **An optimal cutoff at modulus 97.** For

$$G(q) = Q(q^5, q^{97})Q(q^4, q^{97})Q(q^9, q^{97})Q(q^{13}, q^{97}) = \sum_{n \ge 0} g(n)q^n,$$

the four known infinite vanishings arise from the reflection family $(m-n, n, m, m+n)$ with $(m, n) = (9, 4)$.

Theorem 6 (Complete classification at 97)**.** *For $0 \le R < 97$,*

$$(\forall N \ge 0,\ g(97N + R) = 0) \quad \Longleftrightarrow \quad R \in \{9, 13, 55, 64\}.$$

Every other residue has a nonzero coefficient with $N \le 15$. Moreover $g(97N+54) = 0$ for $0 \le N < 15$, and $g(1509) = 1$. Consequently the common cutoff 15 is optimal for this product.

Proof. The reflection theorem proves the four infinite zero progressions. For each of the other 93 residues, a generated exact coefficient certificate proves nonvanishing by $N = 15$. All required degrees are at most 1509. Nonnegativity of the four Watson exponents and their quadratic growth bound every contributing integer index in $[-3, 3]$. The resulting finite coefficient sums, the 15 preceding zero coefficients in residue 54, and the nonzero value at 1509 are evaluated by Lean's kernel. The Python generator only proposes certificates. □

Theorem 5 supplies the necessity that the earlier sixteen-map compatibility calculation alone did not establish. The classification of arbitrary four-tuples and composite-modulus necessity remain open here. A uniform quarter-prime first-witness bound cannot be used for four factors: the repository proves counterexamples, including one inside the reflection family at modulus 2521. The triple quarter-prime conjecture is a separate unresolved quantitative question.

6. Formal verification and review guide

The latest full-root proof snapshot is repository commit `012eb722b2c3a65ec9e8857e84d2611abb4133d6`, with Lean `4.30.0-rc2` and the dependencies pinned by `lake-manifest.json`. The 10,597-job full build and 1,087 distinct audited declarations are recorded in `docs/verification/four_prime_converse.json`; their axiom dependencies are confined to `propext`, `Classical.choice` and `Quot.sound`. The earlier triple-review record `docs/verification/quintuple_review.json` is a historical snapshot. The updated manuscript and documentation inventory are recorded separately in `docs/verification/documentation_2026_11_12.json`. An axiom audit is accompanied by statement and definition checks; axiom cleanliness by itself does not establish that a theorem states the intended mathematical claim.

The principal source map, with paths relative to `RamanujanTau/`, is:

Mathematical step	Formal module (prefix `MultiQuintuple`)
Actual products and root predicate	`RootVanishingClassification.lean`
Poisson bridge and shell cancellation	`DualGaussianCancellation.lean`
Phase normalization and antipodes	`CanonicalPhaseCharacter.lean`
Opposite supported partners	`DualShellPartners.lean`
Integral short generation	`UniversalShortGeneration.lean`
Exceptional modulus 13	`ConverseThirteen.lean`
Arithmetic extraction of the root	`ShortDualConverse.lean`
Final equivalence	`FullConverse.lean`
Centered-root completeness	`CenteredRoots.lean`
Four-factor converse at 97	`FourNinetySevenConverse.lean`
Four-dimensional phase partners	`FourPhaseParity.lean`
Integer generator certificates	`FourPrimeIntegral.lean`
Two planes and character exclusions	`FourPrimeBlockCharacters.lean`
Phase and small-prime completion	`FourPrimePhaseBlocks.lean`, `FourPrimeSmallCases.lean`, `FourPrimeSmall89.lean`
Prime-family final equivalence	`FourPrimeConverse.lean`

Run the reproduction script from the repository root:

```
python3 scripts/verify_quintuple_review.py
python3 scripts/verify_lean.py --full
python3 scripts/check_four_prime_converse.py --self-test
python3 scripts/generate_four_prime_converse.py --check
python3 scripts/generate_four_prime_small.py --check
python3 scripts/classify_sparse_triple.py 31 1 5 6
```

The first command builds the relevant modules, checks the dedicated axiom audit, independently replays the finite lattice certificates, compares the reference root scan with direct products on a small exhaustive census, and checks the generated modulus-97 source and coefficient witnesses. It writes a verification record and logs. The public verifier with `-full` builds the entire root and runs the four public/headline audits. The prime-family checker independently replays the integral and finite obligations, including five corruption rejection controls; the generators check that their output reproduces the saved certificates.

Independent review should concentrate on the exact product convention, the normalization of the Fourier phase, passage from shell cancellation to opposite partners, integral generation rather than real spanning, the exhaustive normalized coverage below 1250, and extraction of the two triple residue equations. For the four-factor theorem, review should also check the missing-phase argument, integral generation, preservation of the two planes, modulo-nine block exclusions, and small-prime coverage. The literature comparison must also account for the further work announced in [2].

Assistance and publication status. This draft and the formal development were prepared with AI assistance. The draft has not undergone independent human mathematical review or peer review. It is prepared for that review; no journal submission, external correspondence or priority claim is implied.

References

[1] T. Daniels, T. Huber, J. McLaughlin and D. Ye, *The p-Dissection of a Product of Quintuple Products*, arXiv:2603.04666 (2026), https://arxiv.org/abs/2603.04666.

[2] T. Daniels, T. Huber, J. McLaughlin and D. Ye, *Vanishing Coefficients in Products of Quintuple Products*, arXiv:2606.06701 (2026), https://arxiv.org/abs/2606.06701.

[3] S. Chern and D. Tang, *General coefficient-vanishing results associated with theta series*, Advances in Applied Mathematics **159** (2024), 102742, https://arxiv.org/abs/2208.09882.

[4] T. DiFiore, *RamanujanLean*, source and verification records, https://github.com/tomdif/RamanujanLean.